Raksha Kumaraswamy

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RESEARCH INTERESTS

Reinforcement learning, representation learning, machine learning.

EMPLOYMENT

Senior Research Scientist, *Sony AI* — 2023 – present
Working in a small team to build RL-based AI agents for PlayStation games — Toronto, ON, Canada

Senior Researcher, *Huawei Technologies Canada Co. Ltd.* — 2021 – 2023
Worked on research projects related to fundamental RL like exploration, multi-agent learning (for NN training) — Edmonton, AB, Canada

EDUCATION

Ph.D., Computing Science, *University of Alberta* — 2017 – 2021
Transferred from Indiana University. Co-advisors: Martha White, Adam White — Edmonton, AB, Canada

Ph.D. studies, Computer Science, *Indiana University Bloomington* — 2015 – 2017
Advisor: Martha White — Bloomington, IN, USA

M.S., Computer Science, *Indiana University Bloomington* — 2013 – 2015
Advisor: Martha White — Bloomington, IN, USA

B.E., Computer Science, *S. J. B. Institute of Technology, Visvesvaraya Technological University* — 2008 – 2012
Karnataka, India

RESEARCH EXPERIENCE

Research Assistant, *RLAI Lab, University of Alberta* — 2017 – 2021
Reinforcement Learning and Artificial Intelligence Lab. Advisor: Martha White — Edmonton, AB, Canada

Research Assistant, *Machine Learning Lab, Indiana University* — 2015 – 2017
Advisor: Martha White — Bloomington, IN, USA

Research Assistant, *StaR-AI Lab, Indiana University* — 2013 – 2015
Advisor: Sriraam Natarajan — Bloomington, IN, USA

Research Assistant, *Game Theory Lab, Indian Institute of Science* — Winter 2013
Advisor: Y. Narahari — Bangalore, India

PUBLICATIONS

JOURNAL ARTICLES

- **Investigating the Properties of Neural Network Representations in Reinforcement Learning** [Paper]
  *H. Wang, E. Miahi, M. White, M. C. Machado, Z. Abbas, **R. Kumaraswamy**, V. Liu, A. White*
  In *Artificial Intelligence Journal (AIJ)*, 2024.
- **Off-Policy Actor-Critic with Emphatic Weightings** [Paper]
  *E. Graves, E. Imani, **R. Kumaraswamy**, M. White*
  In *Journal of Machine Learning Research (JMLR)*, 2023.
- **From Eye-Blinks to State Construction: Diagnostic Benchmarks for Online Representation Learning** [Paper]
  *B. Rafiee, Z. Abbas, S. Ghiassian, **R. Kumaraswamy**, R. S. Sutton, E. A. Ludvig, A. White*
  In *Adaptive Behavior*, 2023.

CONFERENCE PAPERS

- **Value Bonuses using Ensemble Errors for Exploration in Reinforcement Learning** [Paper]
  *A. Wahab, **R. Kumaraswamy**, M. White*
  In *Reinforcement Learning Conference (RLC)*, 2025.
- **The Cross-Environment Hyperparameter Setting Benchmark for Reinforcement Learning** [Paper]
  *A. Patterson, S. Neumann, **R. Kumaraswamy**, M. White, A. M. White*
  In *Reinforcement Learning Conference (RLC)*, 2024.
- **Continual Auxiliary Task Learning** [Paper]
  *M. McLeod, C. Lo, M. Schlegel, A. Jacobsen, **R. Kumaraswamy**, M. White, A. White*
  In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- **The Utility of Sparse Representations for Control in Reinforcement Learning** [Paper]
  *V. Liu, **R. Kumaraswamy**, L. Li, M. White*
  In *AAAI Conference on Artificial Intelligence (AAAI)*, 2019.
- **Context-Dependent Upper-Confidence Bounds for Directed Exploration** [Paper]
  ***R. Kumaraswamy**, M. Schlegel, A. White, M. White*
  In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- **Learning Sparse Representations in Reinforcement Learning with Sparse Coding** [Paper]
  *L. Le, **R. Kumaraswamy**, M. White*
  In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2017.
- **Learning through Advice-Seeking via Transfer** [Paper]
  *P. Odom, **R. Kumaraswamy**, K. Kersting, S. Natarajan*
  In *International Conference on Inductive Logic Programming (ILP)*, 2016.
- **Transfer Learning via Relational Type Matching** [Paper]
  ***R. Kumaraswamy**, P. Odom, K. Kersting, D. Leake, S. Natarajan*
  In *IEEE International Conference on Data Mining (ICDM)*, 2015.

WORKSHOP PAPERS

- **Transfer Learning Across Relational and Uncertain Domains: A Language-Bias Approach**
  ***R. Kumaraswamy**, P. Odom, K. Kersting, D. Leake, S. Natarajan*
  In *International Workshop on Statistical Relational AI (StarAI)*, 2015.

POSTERS

- **Learning Sparse Representations in Reinforcement Learning with Sparse Coding**
  ***R. Kumaraswamy**, L. Le, M. White*
  In *Multidisciplinary Conference on Reinforcement Learning and Decision Making (RLDM)*, 2017.
- **Anomaly Detection in Text: The Value of Domain Knowledge** [Paper]
  ***R. Kumaraswamy**, A. Wazalwar, T. Khot, J. Shavlik, S. Natarajan*
  In *International FLAIRS Conference (FLAIRS)*, 2015. **Best Poster Award.**

TEACHING EXPERIENCE

CMPUT 466/566: Machine Learning, *University of Alberta* Fall 2019

CMPUT 366: Intelligent Systems, *University of Alberta* Fall 2018

CSCI B555: Machine Learning, *Indiana University Bloomington* Fall 2016

SERVICE

Organizer	RLC Workshops Chair (2026)
Reviewer	ICLR, ICML, NeurIPS (2019 – present); RLC (2024 – present); AAAI (2019)
Department	Computer Science Colloquium Committee, Indiana University (2016 – 2017)
University	Events Chair, Women in Computing, Indiana University (2013 – 2014)